

Education of Children Left Behind: Evidence from Migrant-Sending Households in Bangladesh

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Abstract

Migration plays a central role in household strategies in rural Bangladesh, offering financial gains through remittances but potentially disrupting children's education due to parental absence. This study examines the impact of household migration on school enrollment of children left behind. We use panel data from the Bangladesh Integrated Household Survey (BIHS) covering 2012, 2015, and 2018, comprising 11,739 child observations across three waves. The BIHS provides comprehensive information on household demographics, school enrollment, migration patterns, and economic characteristics, making it suitable for this study. Empirical methods include both standard Difference-in-Differences (DiD) and Fuzzy DiD models, accounting for potential endogeneity of migration by instrumenting with predicted migration probabilities from fixed-effects logit models. The findings reveal that in the 2012–2015 period, external migration significantly reduced child enrollment, while internal and parental migration had limited effects. In the 2015–2018 period, overall household migration was associated with decreased enrollment. These results highlight the dual impact of migration, financial gains through remittances versus educational disruption from parental absence, and underscore the need for policies that support children in migrant households.

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Contents

1	Introduction	3
2	Literature Review	4
3	Theoretical Framework	8
3.1	Human Capital Approach	8
3.2	New Economics of Labor Migration (NELM)	8
3.3	Social Remittances	10
3.4	Community Spillovers	11
4	Research Methods	13
4.1	Data Source	13
4.2	Variable Construction	13
4.3	Rationale for the Method	16
4.4	Baseline Model: Standard DiD	16
4.5	Identification Strategy: Fuzzy DiD with Instrumental Variables	17
4.6	Treatment Definitions: Dimensions of Migration	18
5	Empirical Findings	19
5.1	Difference-in-Differences (DiD) Estimates	19
5.2	Fuzzy Difference-in-Differences (Fuzzy DiD)	20
6	Discussion and Interpretation of Findings	21
6.1	Interpretation of DiD Results	22
6.2	Interpretation of Fuzzy DiD Results	22
6.3	Mechanisms and Theoretical Insights	23
6.4	Limitations of the Study	23
7	Conclusion	24

1 Introduction

In Bangladesh, migration is not an exception but a common household strategy, especially among rural families facing economic vulnerability. For many, farming alone is not enough to sustain livelihoods, and households regularly confront shocks such as floods, cyclones, and crop failures. In such contexts, sending one family member to work in cities or abroad has become both a survival strategy and an investment decision. Official statistics show that more than 13 million Bangladeshis have migrated internationally, primarily to the Gulf States, Malaysia, and in recent years to Europe and Southeast Asia. At the same time, internal migration to Dhaka, Chattogram, and other district towns has become increasingly visible, driven by underemployment in rural areas and the attraction of urban labor markets.

Households generally view migration as more than just a source of immediate income. Migration earnings are often invested in long-term goals, such as purchasing land, constructing houses, or financing children's education. Remittances sent back by migrant family members can help pay school fees, buy books, and even fund private tuition, improving children's access to education. These financial flows can ease household budget constraints and offer opportunities that might otherwise be unattainable in low-income rural settings.

At the same time, migration introduces challenges, particularly for children left behind. Parental absence may increase children's household responsibilities, requiring them to care for younger siblings, help with domestic chores, or support family farming activities. Such increased responsibilities can limit the time and energy available for studying, and in some cases, children may experience emotional strain due to separation from parents. The combined effect of migration on education is thus complex: while financial benefits may support schooling, the absence of parents and the associated household adjustments may hinder children's educational participation.

The effects of migration are also shaped by the type and destination of migration. International migration may bring higher and more regular remittances, while internal

migration, particularly to nearby urban areas, may involve shorter absences and different household adjustments. Additionally, social and cultural influences play a role. Migrants often transmit ideas, skills, and aspirations—sometimes referred to as social remittances—which can influence household decision-making and children’s educational ambitions. Communities with high migration rates may experience spillover effects, where both economic gains and perceptions of migration as a desirable livelihood affect schooling choices. Gender may further mediate these effects, as boys and girls respond differently to migration-related household and community dynamics.

Despite the potential importance of migration for children’s education, empirical evidence in Bangladesh remains limited. Most studies focus on economic impacts or remittance flows, while the educational outcomes of children left behind receive less attention, particularly in a causal framework. Understanding this relationship is critical because education shapes long-term human capital development, income potential, and social mobility. It is also important for policy, as programs aimed at supporting migrant households or children left behind must consider both economic and social dimensions of migration.

In this study, we examine the effect of household migration on the education of children left behind in Bangladesh. Using household survey data and employing both standard Difference-in-Differences (DiD) and Fuzzy DiD methods, we explore how internal and international migration, as well as parental migration, influence children’s school enrollment. We account for household characteristics, economic conditions, and other relevant factors to isolate the potential causal effect of migration.

This research addresses a critical question: *Does the migration of household members affect the education of children left behind in Bangladesh? If so, how?*

2 Literature Review

It is now well accepted that international migration of a parent or family member can have both positive and negative effects on children who remain in the home country. On the positive side, remittances relax household budget constraints, making it possible to

increase child schooling, improve health, and reduce child labor. Several studies provide supporting evidence (Cox-Edwards & Ureta, 2003; Alcaraz et al., 2012; Yang, 2008). At the same time, parental migration inherently means parental absence, which can negatively affect child outcomes and may even outweigh the benefits of remittances.

Evidence from different countries shows this tension. For example, Giannelli and Mangiavacchi (2010) find that parental migration reduced school attendance of children in Albania. In Mexico, Lahaie et al. (2009) document that children with a migrant parent experienced academic, behavioral, and emotional difficulties. Zoller Booth (1995) stressed the role of fathers as disciplinarian and figurehead in Swaziland, and showed how paternal migration harmed school readiness of children. Similar evidence is found in Mexico, where Antman (2011a) identify problems of father absence linked to education.

Gender and age of children also matter. In China, Meyerhoefer and Chen (2011) show that girls' education suffered when fathers migrated, because their time shifted toward household production. In Mexico, McKenzie and Rapoport (2011) find that older children left behind are pulled into housework (girls) or early migration (boys). By contrast, Acosta (2011) find that in El Salvador, remittances reduced child wage labor, with girls benefitting more in schooling than boys.

The gender of the migrant parent is another dimension. In the Philippines, Cortes (2010) shows that maternal migration had a negative effect on children's education, while paternal migration did not. Similar results are found in Thailand, where long-term maternal absence reduced children's educational outcomes (Jampaklay, 2006). These findings suggest that maternal absence may be more detrimental than paternal absence.

Migration can also change bargaining power within the household. If men migrate, women may gain more decision-making power. In Mexico, Antman (2011b) show that when fathers migrate, families spent relatively more resources on girls compared to boys. Antman (2010) uses panel data to show that such shifts are temporary: while men are away, girls benefit more; when fathers return, the pattern reverses. This is consistent with broader findings that women's bargaining power tends to favor girls in health and education

(Thomas, 1994; Duflo, 2003).

Another mechanism is the “brain gain” hypothesis: when migration prospects are high, even children who do not migrate may study more, to prepare for possible migration. The evidence here is mixed. Boucher et al. (2005) find no effect of international migration on schooling investment in Mexico, while deBrauw and Giles (2006) show that internal migration opportunities in China reduced high school enrollment. In the U.S.-Mexico context, Kandel and Kao (2000) show that children of migrants often had lower aspirations for schooling, as migration appeared a quicker path to mobility. In Cape Verde, by contrast, Batista et al. (2007) found that migration prospects increased secondary school completion. These effects depend on the home and destination country pairing, and the imperfect transferability of educational attainment across borders (Gibson & McKenzie, 2011; Beine et al., 2011).

Finally, timing is important. Migration may have different effects in the short term and long term. In Mexico, Antman (2011c) shows that paternal migration initially reduced boys’ study hours within the first year, when remittances were not yet received. But in the long run, paternal migration increased educational attainment of girls (Antman, 2011a). Similarly, Hanson and Woodruff (2003) show that Mexican girls benefitted in terms of completed schooling when living in households with external migrants.

Turning to Bangladesh, recent work highlights similar complexities. Ahmed et al. (2024) show that parental internal migration boosts income and asset accumulation in ultra-poor households, but reduces children’s educational enrollment and spending, likely due to lack of parental care. Complementary evidence from Khanam et al. (2023) suggests that parental migration—particularly when both parents migrate—substantially lowers the likelihood of favorable early childhood development outcomes. In India, Kusuma and Babu (2024) similarly report that children of poor migrants had lower levels of schooling compared to those living in their native places.

Studies have investigated the determinants of migration in Bangladesh, both internal and international. Islam et al. (2021) analyze household-level data from the Bangladesh

Household Income and Expenditure Survey (HIES) 2010 to identify key factors influencing migration decisions. They find that the age of the household head, farm size, household assets, number of young dependents, and district-level economic opportunities significantly affect the likelihood of migration. In particular, larger farm size and higher asset values increase both internal and external migration probabilities. Households with loans are more likely to undertake internal migration, while external migration is more common overall, particularly from rural areas. These findings indicate that migration in Bangladesh is shaped by household socio-economic characteristics, local economic opportunities, and credit constraints, providing an important context for understanding how migration might affect children left behind.

Overall, the literature shows positive, negative, and even null effects of parental migration on children’s education depending on context, survey design, and timing of measurement. Positive impacts highlight remittances reducing liquidity constraints, while negative impacts emphasize parental absence, shifting responsibilities, and psychosocial costs. Gender differences and exposure duration further explain divergent findings. These insights guide our analysis for rural Bangladesh, where both international and internal migration are widespread but context-specific.

Contribution of this study: While prior Bangladesh-based studies provide important insights, they focused on short-term effects or specific regions and do not distinguish between internal and international migration, or parental versus other household member migration. This study contributes to the literature by analyzing both internal and external migration, differentiating parental and non-parental migration, and employing quasi-experimental methods (DiD and Fuzzy DiD) to examine causal effects on children’s school enrollment across multiple periods. This approach allows a more nuanced understanding of how migration shapes educational outcomes for children left behind in rural Bangladesh.

3 Theoretical Framework

This section outlines the theoretical perspectives that guide the analysis of how migration affects children’s education. Four complementary frameworks are discussed. Together, these frameworks provide a multidimensional lens for understanding the complex relationship between migration and educational outcomes.

3.1 Human Capital Approach

The human capital approach posits that households make decisions about children’s education by weighing the expected costs and benefits. Benefits include the prospect of better future jobs, higher earnings, and social mobility, while costs comprise tuition fees, books, and the opportunity cost of children contributing to household labor. Migration alters this cost-benefit calculation in multiple ways. On the one hand, remittances from migrant family members reduce the financial burden of schooling, making education more affordable and accessible. On the other hand, the absence of parents due to migration can increase children’s responsibilities at home, potentially pushing them into work and reducing the time and attention they can devote to study.

As illustrated in Figure 1, migration functions as a shock to the household decision-making process. It simultaneously enhances the benefits side through increased household income and imposes additional costs due to parental absence and the resulting labor demands on children. Consequently, the final outcome for children’s education is context-dependent: in some households, schooling is maintained or even improved due to increased resources, while in others, children may drop out or reduce their educational participation due to higher domestic responsibilities. This framework highlights the nuanced and dual influence of migration on educational outcomes through the lens of human capital theory.

3.2 New Economics of Labor Migration (NELM)

The New Economics of Labor Migration (NELM) framework conceptualizes migration as a *household-level strategy* rather than merely an individual decision. In contexts where

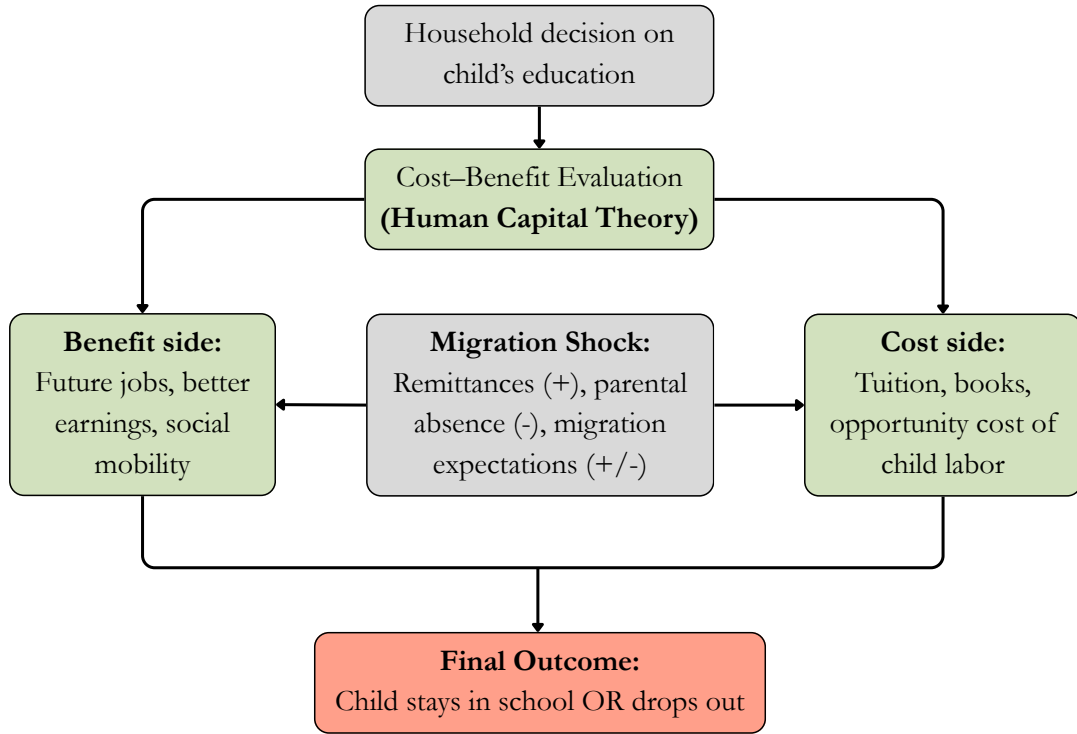


Figure 1: Human Capital Theory Framework

credit and insurance markets are weak, households use migration to diversify income sources and protect themselves against economic shocks. This strategy has important implications for children's education. On the positive side, remittances can relax household budget constraints, secure tuition payments, and provide a form of insurance during poor harvests or unexpected family emergencies, thereby protecting children's schooling. However, migration also has potential negative effects: the absence of a parent may increase the labor burden on children or disrupt their study routines, particularly if households rely on children to compensate for lost labor.

At the same time, migration decisions are often influenced by aspirations for children's education. Households may send members abroad specifically to create opportunities for their children to access better schooling or training. As a result, the net outcome for child education is context-dependent and may vary across households and communities: for some children, schooling is effectively protected or enhanced, while for others, the disruption caused by parental absence or household adjustments may reduce educational

participation. Figure 2 visually represents this interplay of economic protection, potential risks, and educational aspirations within the NELM framework.

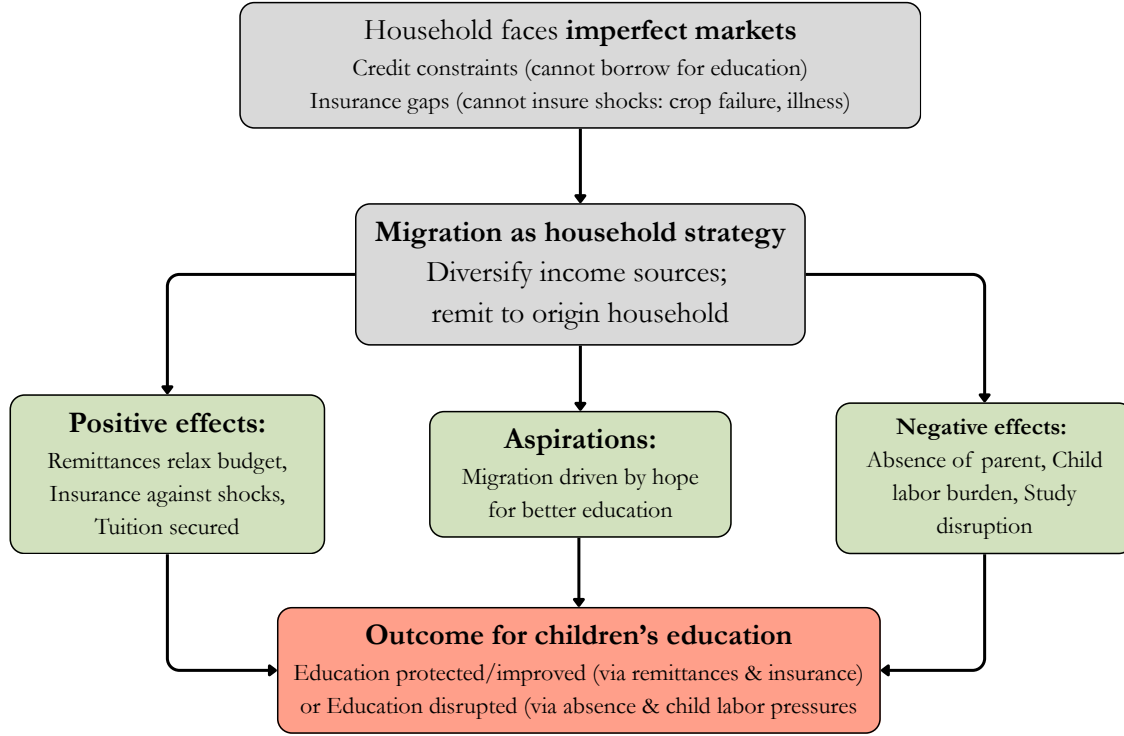


Figure 2: NELM Theoretical Framework

3.3 Social Remittances

Social remittances refer to the *ideas, norms, skills, and practices that migrants transmit back to their home communities*, alongside financial remittances. This concept highlights how migration can generate *non-monetary spillovers*, shaping social behavior, cultural norms, and household decision-making in origin communities. These remittances often influence children's aspirations, family strategies, and broader community practices, creating a pathway for social and educational change that goes beyond purely economic effects.

The mechanism through which social remittances affect children and communities is illustrated in Figure 3. When migration occurs, household members send back ideas, values,

and experiences, which constitute social remittances. The nature of these remittances depends on the migration context. For households with members migrating to the Gulf, families often emphasize *job-oriented training* for children, reflecting labor market demands in the destination countries. In contrast, households with members migrating to Western countries tend to prioritize *formal schooling*, anticipating long-term educational and professional benefits. Over time, these social remittances influence children’s aspirations and gradually reshape community-level norms regarding education and training, creating a complex mix of outcomes that vary by household, gender, and local context.

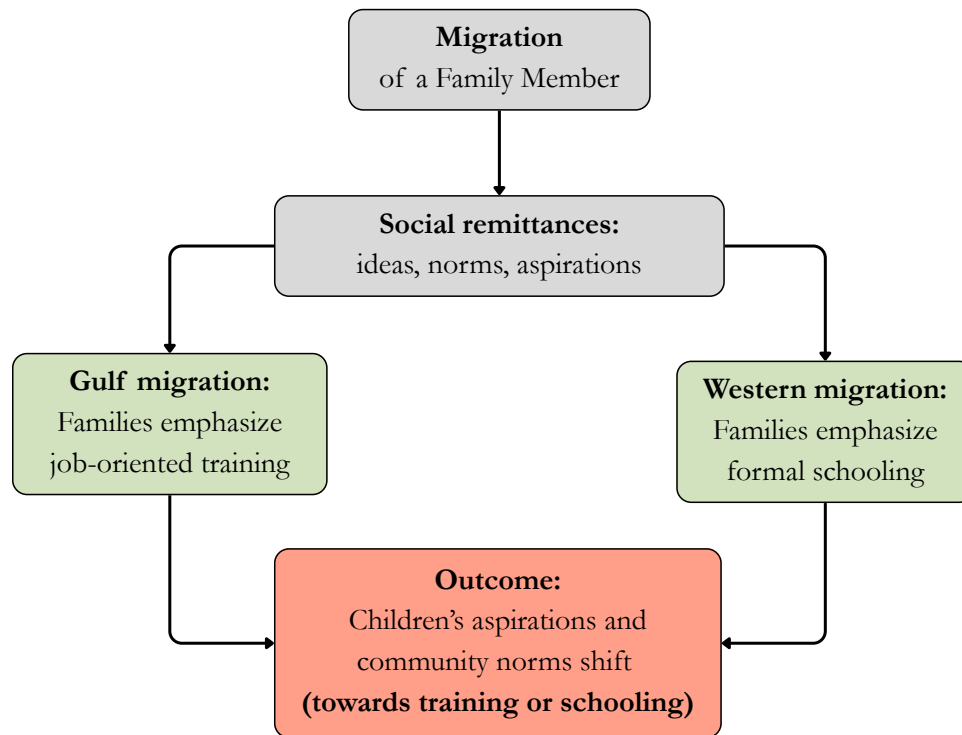


Figure 3: Social Remittances Theoretical Framework

3.4 Community Spillovers

Spillover theory in migration studies examines how the behavior or outcomes of some households or individuals affect others in the same community, beyond direct recipients of migration benefits. In the context of child education, high levels of migration in a village can produce both economic and social spillovers.

The Figure 4 illustrates the dual pathways through which high migration in a village affects children’s schooling. The economic effect of migration increases household income through remittances or higher local wages, which reduces the need for child labor and encourages school attendance. In contrast, the social and cultural effect arises from migration being perceived as glamorous, especially among boys, who may then prefer migration over schooling. Girls are generally less influenced by this social aspiration, often maintaining or slightly increasing their school participation. The interplay of these two effects generates a mixed overall outcome: while some children benefit from improved economic conditions and attend school more regularly, others may reduce their schooling or drop out due to the perceived attractiveness of migration as a livelihood strategy. This framework highlights the complex, gender-sensitive, and context-dependent nature of migration’s impact on educational outcomes at the community level.

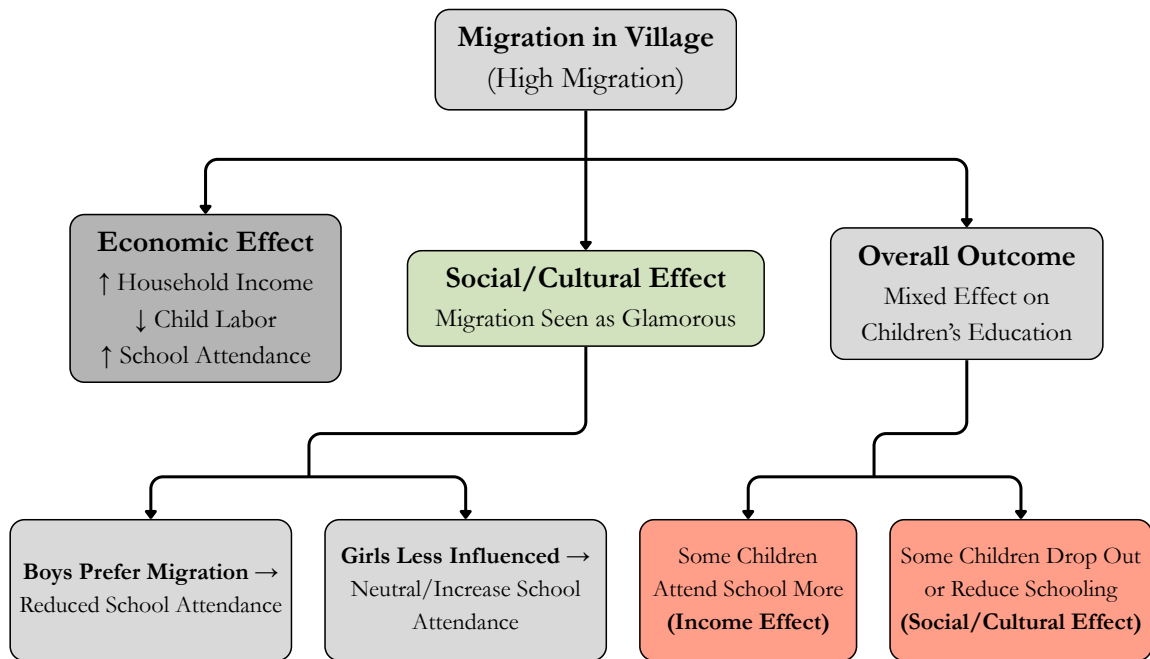


Figure 4: Community Spillovers Theoretical Framework

4 Research Methods

4.1 Data Source

This study uses data from the Bangladesh Integrated Household Survey (BIHS), a nationally representative panel survey conducted by the International Food Policy Research Institute (IFPRI). The BIHS covers rural households across all seven administrative divisions of Bangladesh and includes detailed information on school enrollment, migration, household characteristics etc.

The survey employs a two-stage stratified sampling design, selecting primary sampling units (PSUs) followed by random household selection within PSUs. Detailed data on migrant household, migration period, migration destination, school enrollment, allow for construction of treatment and outcome variables. The study considered all three waves of the dataset.

- Round I: 2012 (6,500 households)
- Round II: 2015 (6,500 households)
- Round III: 2018 (5,604 households)

Attrition between 2012 and 2015 was only 4.4%. The sample includes 11,739 children observations across the three waves. Based on these information the BIHS dataset was considered an appropriate dataset for the empirical exercise.

4.2 Variable Construction

All the following variables were curated from the Bangladesh Integrated Household Survey (BIHS) across its three waves, using information from multiple survey modules. This setup allows the study to estimate the causal impact of migration on child education while accounting for household resources, child characteristics, and community context. The main outcome of this study is **child school enrollment**, which measures whether a child is attending school.

- **School enrollment:** measured as whether the child is enrolled in any school.

Setup: binary variable; 1 if not enrolled, 0 if enrolled.

- **Type of school:** indicates whether the child attends a government school.

Setup: binary variable; 1 for government school, 0 for non-government school.

- **Household income:** captures the economic resources of the household.

Setup: log-transformed continuous variable.

To capture the effect of household migration on child education, the study defines several treatment variables.

- **Overall household migration:** indicates if any household member has migrated.

Setup: binary variable; 1 if at least one member migrated, 0 if none migrated.

- **Parental migration:** indicates if at least one parent (father or mother) has migrated.

Setup: binary variable; 1 if a parent migrated, 0 if no parent or only non-parental migration.

- **External migration:** indicates if any household member has migrated abroad.

Setup: binary variable; 1 if at least one member migrated abroad, 0 otherwise.

- **Internal migration:** indicates if any household member has migrated within the country to cities or district towns.

Setup: binary variable; 1 if at least one member migrated internally, 0 otherwise.

To estimate household migration in the first-stage regression, household-level characteristics are used as instruments.

- **Outstanding loan:** measures access to credit for migration.

Setup: log-transformed continuous variable.

- **Total farm size:** measures landholding and resource capacity.

Setup: continuous variable.

- **Household assets:** measures wealth of the household.

Setup: log-transformed continuous variable.

- **Dependency ratio:** ratio of dependent household members to working-age members.

Setup: continuous variable.

- **Predicted migration probability:** predicted likelihood of household migration from first-stage regression.

Setup: continuous variable, used as treatment in fuzzy DiD.

In the main regression estimating the effect of migration on child schooling, several control variables account for child, household, and community characteristics.

- **Distance to school:** proximity to nearest school.

Setup: continuous variable (kilometers).

- **Household size:** total number of household members.

Setup: continuous variable.

- **Village average education expenditure:** captures peer or community influence on schooling.

Setup: continuous variable.

- **Child gender:** gender of the child.

Setup: binary variable; male/female.

- **Child age:** age in years.

Setup: continuous variable.

- **Oldest child indicator:** whether the child is the oldest in the household.

Setup: binary variable; 1 if oldest, 0 otherwise.

This setup allows the study to estimate the causal impact of migration on child education while accounting for household resources, child characteristics, and community context.

4.3 Rationale for the Method

The objective of this paper is to identify whether household migration affects the school enrollment of children who are left behind in Bangladesh. A simple comparison between migrant and non-migrant households may not reflect the true causal effect, because these two groups are systematically different. For instance, poor rural families facing shocks like floods and cyclones are more likely to send a migrant, while households with higher education and social networks may also be overrepresented among migrants. If these pre-existing differences are not controlled, the estimate of migration effect on education will be biased.

To address this issue, the paper applies a Difference-in-Differences (DiD) strategy. The main idea of DiD is to compare the change in school enrollment for children in migrant households before and after migration, with the change for children in non-migrant households over the same period. By differencing across time and groups, DiD removes time-invariant unobserved heterogeneity between households, as well as common time shocks such as policy changes or macroeconomic fluctuations.

4.4 Baseline Model: Standard DiD

Standard DiD was used to estimate the impact of migration, which is binary treatment. The considered variables are:

- $NotEnroll_{it}$: binary indicator (1 if child i in household h is enrolled in school in year t , 0 otherwise),
- Mig_h : treatment indicator (1 if household h has at least one member migrated, 0 otherwise),
- $Post_t$: time dummy (1 if after migration year, 0 if before),
- X_{it} : vector of control variables including child age, child gender, household head education, household asset index, household size, and exposure to shocks,
- ϵ_{it} : error term.

The baseline DiD model is specified as:

$$NotEnroll_{it} = \alpha + \beta(Mig_h \times Post_t) + \gamma Mig_h + \delta Post_t + \theta X_{it} + \epsilon_{it} \quad (1)$$

Here, β captures the *average treatment effect on the treated (ATT)*. It measures the effect of household migration on the probability of school enrollment of children left behind.

4.5 Identification Strategy: Fuzzy DiD with Instrumental Variables

Migration decision is not random. Households may choose to migrate based on unobserved characteristics that also influence children's schooling, such as parental motivation, ambition, or access to informal credit. This makes treatment assignment endogenous and causes bias in the standard DiD estimate.

To address this concern, a fuzzy DiD approach is applied using instrumental variables (IV). The logic is similar to two-stage least squares (2SLS), where the first stage predicts the probability of migration, and the second stage uses this predicted probability in the DiD regression.

First Stage: Probit Model for Migration

$$Mig_{ht}^* = Z_{ht}\pi + u_{ht} \quad (2)$$

where Mig_{ht}^* is the latent migration decision, Z_{ht} is a vector of instruments (historical migration rate at the village level, distance to manpower recruiting agency, exposure to floods and cyclones), and u_{ht} is the error term.

From this model, the fitted migration probability is constructed as:

$$\hat{Mig}_{ht} = Pr(Mig_{ht} = 1 \mid Z_{ht}) \quad (3)$$

Second Stage: Fuzzy DiD Regression

$$NotEnroll_{it} = \alpha + \beta(\hat{Mig}_{ht} \times Post_t) + \theta X_{it} + \epsilon_{it} \quad (4)$$

Here, β captures the causal impact of household migration on child school enrollment. By using $\hat{Mig}_{ht}$ rather than actual Mig_{ht} , bias from endogenous household decisions is reduced.

4.6 Treatment Definitions: Dimensions of Migration

In addition to migration as a whole, treatment is differentiated into several categories to capture heterogeneity. Each treatment variable is defined as a binary indicator:

a) Overall migration vs. no migration

$Mig_h = 1$ if household has at least one migrant (internal or external)

$Mig_h = 0$ if household has no migrant

b) Internal migration vs. no migration

$IntMig_h = 1$ if at least one member migrated to major cities

$IntMig_h = 0$ if household has no migrant

c) External migration vs. no migration

$ExtMig_h = 1$ if at least one member migrated abroad

$ExtMig_h = 0$ if household has no migrant

d) Parental migration vs. all others

$ParMig_h = 1$ if a parent (father or mother) migrated

$ParMig_h = 0$ if household has no migrant or only non-parental migrant

For each case, both standard DiD and fuzzy DiD estimates are obtained. This allows testing whether the effect of migration differs across types of migration and role of the migrant.

The empirical strategy proceeds in three steps. First, a standard DiD model estimates the baseline effect of migration on school enrollment. Second, a fuzzy DiD framework with IV corrects for potential endogeneity. Third, treatment is disaggregated into internal vs. external migration, and parental migration vs. all others. This provides robust and nuanced estimates of the causal impact of migration on education outcomes in Bangladesh.

5 Empirical Findings

5.1 Difference-in-Differences (DiD) Estimates

Table 1 presents the ATET estimates from standard DiD models for the 2012–2015 period. We report estimates for overall household migration, internal migration, external migration, and parental migration, controlling for household and child characteristics.

Table 1: DiD Estimates of Migration on Child Enrollment (2012–2015)				
	Migration	External Migration	Internal Migration	Parent Migration
ATET	0.006	-0.219***	0.033	-0.011
	(0.107)	(0.071)	(0.034)	(0.016)
Controls	Yes	Yes	Yes	Yes
Observations	4306	3904	4048	4306

Interpretation: Overall household migration has no significant effect on enrollment during 2012–2015. External migration negatively affects enrollment, while internal and parental migration are not statistically significant. Controls indicate that attending a government school strongly increases the likelihood of enrollment, gender differences are significant (male children slightly less likely), and household characteristics such as income, size, distance to school, and village average education show smaller or insignificant effects.

Full results for controls are reported in the Annex (Tables A1–A2).

Table 2 presents DiD estimates for 2015–2018.

Table 2: DiD Estimates of Migration on Child Enrollment (2015–2018)				
	Migration	External Migration	Internal Migration	Parent Migration
ATET	-0.239*** (0.053)	-0.114*** (0.031)	-0.274*** (0.061)	0.042* (0.024)
Controls	Yes	Yes	Yes	Yes
Observations	3774	3439	3656	3774

Interpretation: Overall migration is associated with a significant decline in enrollment in 2015–2018. Both internal and external migration negatively affect enrollment, while parental migration shows a small positive effect. Control variables continue to show that government school attendance strongly increases enrollment, gender and age remain significant, and distance to school negatively affects enrollment. Household income shows a stronger positive effect compared to the previous period. Full control variable estimates are provided in the Annex (Tables A3–A4).

5.2 Fuzzy Difference-in-Differences (Fuzzy DiD)

To account for the endogeneity of migration, we use predicted migration probabilities from a fixed-effects logit model as instruments and estimate fuzzy DiD.

Table 3 shows the LATE estimates for 2012–2015.

Interpretation: Internal migration shows no significant effect. External migration positively affects enrollment, suggesting remittances or household support benefits. Parental migration significantly reduces child enrollment, highlighting the importance of parental presence. Control variables show patterns similar to the standard DiD results: govern-

Table 3: Fuzzy DiD Estimates of Migration on Child Enrollment (2012–2015)

Migration Type	Coefficient	Std. Error
Internal Migration	-1.214	4.730
External Migration	2.780***	1.160
Parental Migration	-0.964***	0.259

Notes: * p $\leq$ 0.10, ** p $\leq$ 0.05, *** p $\leq$ 0.01.

ment school attendance and child gender remain important determinants, while other household characteristics are less influential.

Table 4 presents fuzzy DiD estimates for 2015–2018.

Table 4: Fuzzy DiD Estimates of Migration on Child Enrollment (2015–2018)

Migration Type	Coefficient	Std. Error
Internal Migration	1.697	7.225
External Migration	2.407	19.001
Parental Migration	10.502	44.387

Notes: * p $\leq$ 0.10, ** p $\leq$ 0.05, *** p $\leq$ 0.01.

Interpretation: No migration type is statistically significant in this period. The large standard errors suggest high variability and possibly weaker instruments, making it difficult to draw conclusive results. The pattern of control variables remains consistent with the earlier period, as detailed in the Annex.

Note: All control variable estimates for DiD are reported in Tables 5 and 6 in the Annex. These include government school attendance, household income, household size, distance to school, village average educational experience, gender, age, and birth order of the child.

6 Discussion and Interpretation of Findings

This study examines the effect of household migration on children’s educational enrollment in Bangladesh, considering both internal and external migration as well as parental absence. The results from DiD and Fuzzy DiD models reveal complex and sometimes con-

trasting effects, which can be better understood using the mentioned theoretical frameworks.

6.1 Interpretation of DiD Results

For the 2012–2015 period, standard DiD estimates show that overall household migration does not significantly affect enrollment. External migration negatively affects enrollment, while internal and parental migration are not statistically significant. These findings can be interpreted through the lens of the human capital approach: when parents migrate externally, children may face increased household responsibilities, reducing time for study. Remittances may partially offset this effect, but in this early period, the cost of parental absence dominates.

For the 2015–2018 period, overall migration is associated with a significant decline in enrollment. Both internal and external migration show negative effects, while parental migration exhibits a small positive effect. This shift may reflect household adaptation strategies as described by the New Economics of Labor Migration (NELM) framework: households may increasingly rely on remittances to cover educational costs, and internal migration may no longer strongly disrupt schooling because children are older or better supported by other family members.

6.2 Interpretation of Fuzzy DiD Results

Fuzzy DiD estimates, which account for the endogeneity of migration, reveal additional insights. In 2012–2015, parental migration significantly reduces child enrollment, emphasizing the importance of parental presence, consistent with human capital and NELM perspectives. External migration shows a positive effect, suggesting that remittances or household support can enhance schooling. Internal migration remains largely insignificant.

In 2015–2018, none of the migration types show statistically significant effects in fuzzy DiD models, and standard errors are large. This may indicate weaker instruments or high

heterogeneity across households. It also aligns with community spillover theory: in later years, high migration in villages may create mixed economic and social effects, with some children benefiting from improved household income while others may reduce schooling due to the perceived attractiveness of migration.

6.3 Mechanisms and Theoretical Insights

Several mechanisms explain these observed patterns:

- **Remittances:** Both monetary and social remittances can support children’s schooling by reducing financial constraints and shaping educational aspirations.
- **Parental Absence:** Absence of parents increases household responsibilities for children, sometimes leading to reduced school participation.
- **Community Spillovers:** High migration communities create mixed outcomes, as migration is seen as desirable by some children, while economic gains from remittances promote schooling for others.
- **Household Adaptation:** Over time, families may adjust to migration by reallocating household labor or using remittances more effectively, reducing the negative impact on education.

Overall, the effect of migration is context-dependent, varying by type, timing, and household capacity to cope with parental absence.

6.4 Limitations of the Study

This study faces several limitations that should be considered when interpreting the results.

First, the Fuzzy DiD approach relies on predicted migration probabilities from a fixed-effects logit model as instruments. Diagnostics from the first-stage model indicate that the chosen instruments, household loans, total farm size, household assets, and dependency ratio, are strong and relevant, as they significantly predict household migration. However,

the exogeneity condition is not fully satisfied. This implies that, while the instruments are correlated with the endogenous treatment variable, they may still be correlated with unobserved factors affecting child enrollment. Consequently, the Fuzzy DiD estimates may not fully identify causal effects, particularly in periods or for migration types with large standard errors.

Second, measurement error in household survey data, including recall issues in reporting migration history, remittances, and child schooling, may introduce bias.

Third, although the study draws on theoretical frameworks such as human capital theory, NELM, social remittances, and community spillovers, the empirical models cannot fully capture all mechanisms through which migration affects education. Unobserved household and community factors may still influence the outcomes.

Finally, findings are context-specific to the sampled regions and periods in Bangladesh, limiting generalizability to other contexts with different migration patterns, schooling systems, or labor market structures.

7 Conclusion

This study shows that household migration has nuanced effects on children’s education in Bangladesh. Early external migration negatively affects enrollment due to parental absence, while later effects may be mitigated by remittances and household adaptation. Parental migration is consistently important, with absence reducing enrollment in the short term. Internal migration generally has limited impact.

The findings highlight that both economic and social mechanisms matter: remittances, social remittances, and community spillovers interact to produce complex outcomes. Policy interventions should focus on supporting children left behind, particularly in households with absent parents. Programs that channel remittances toward education, monitor school attendance, and provide local support can reduce negative effects while allowing families to benefit from migration.

In conclusion, migration is a key household strategy in Bangladesh, but its impact on education is not uniform. Understanding the type of migration, parental presence, and household coping strategies is crucial for designing policies that protect children's education and maximize the benefits of migration.

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Annex: Control Variable Estimates

DiD Control Variables (2012–2015)

Table 5: Control Variable Estimates: 2012–2015

Control Variable	Coefficient	Std. Error
Government School	0.183***	0.027
Log Income	0.001	0.033
Distance to School	-0.000	0.000
Household Size	0.015	0.009
Village Average Edu. Exp.	-0.000	0.000
Gender (Male = 1)	-0.093***	0.025
Age of Child	0.023***	0.005
Oldest Child	0.000	0.000

Notes: * p $\leq$ 0.10, ** p $\leq$ 0.05, *** p $\leq$ 0.01.

DiD Control Variables (2015–2018)

Table 6: Control Variable Estimates: 2015–2018

Control Variable	Coefficient	Std. Error
Government School	0.137***	0.031
Log Income	0.036***	0.011
Distance to School	-0.000***	0.000
Household Size	0.014	0.012
Village Average Edu. Exp.	-0.000*	0.000
Gender (Male = 1)	-0.120***	0.032
Age of Child	0.030***	0.004
Oldest Child	0.000	0.000

Notes: * p<0.10, ** p<0.05, *** p<0.01.